

HANRONG ZHANG

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EDUCATION

University of Illinois Chicago

Aug. 2025 – Present

Ph.D. Student in Computer Science [Advisor: Prof. Philip S. Yu](#)

Chicago, USA

- Tencent Project Up Scholarship (4 recipients worldwide)

Zhejiang University

Sep. 2022 – Mar. 2025

MEng Computer Engineering [Ranking: 1/82](#)

Hangzhou, China

- **National Scholarship for Graduate Students** ×2 (Top 0.2%)
- Outstanding Graduate in Zhejiang Province & Zhejiang University

University of Leeds

Sep. 2018 – Jun. 2022

BSc Computer Science [First Class Honours](#)

Leeds, United Kingdom

- Best Student Overall (1/300); Best Student in Computer Science (1/75)
- First-class Full-ride Scholarship (1/75)

Southwest Jiaotong University

Sep. 2018 – Jun. 2022

BEng Computer Science and Technology [Ranking: 1/75](#) [GPA: 92/100](#)

Chengdu, China

- National Scholarship for Undergraduate Students (Top 0.2%)
- Outstanding Graduate in Sichuan Province

INTERNSHIP

Google

May 2026 – Present

Student Researcher, Data Synthesis for Gemini and Long-Horizon GUI Agents

Mountain View, California, US

- Synthesize large-scale, diverse **task queries and GUI trajectories** to strengthen **Gemini's** GUI-agent capabilities.
- Build **Mobile Skills Bench (MSB)**, a benchmark for long-horizon interactive mobile agents. (*Under review at ICLR 2027*)
 - * Synthesize verifiable, multi-stage cross-app tasks, paired with an interactive user simulator.
 - * Benchmark frontier models on long-horizon tasks under both pure-GUI and hybrid (GUI + MCP + CLI) execution.
 - * Design an efficient **skill organization** for mobile agents and quantify the value of textual vs. multimodal skills.
- Build **Android-based mock apps** with resettable backend state and verifiable outcomes for internal Android-agent evaluation.

Alibaba Group

May 2025 – Aug. 2025

Research Intern, RL for Human-Agent Multi-turn Interaction

Hangzhou, Zhejiang, China

- Build scalable simulation environments for multi-turn human-agent tool-call interactions (e.g., retail and airline services), tackling sparse rewards, scarce data, and environment instability.
- Propose a **Tool Dependency Graph** that models tool constraints to synthesize high-quality multi-turn tool-call dialogues with ground-truth trajectories for verifiable rewards; train agents on these trajectories via multi-turn **GRPO**, improving long-horizon tool use in human-agent interaction.

SELECTED PUBLICATIONS AND PREPRINTS

Full publication list is available on [Google Scholar](#). * denotes Equal Contribution. † denotes Corresponding Author.

- [1] **Hanrong Zhang**, Jingyuan Huang, Kai Mei, Yifei Yao, Zhenting Wang, Chenlu Zhan, Hongwei Wang, Yongfeng Zhang, *Agent Security Bench (ASB): Formalizing and Benchmarking Attacks and Defenses in LLM-based Agents*, **ICLR 2025**. [Code] [Website]
[LLM Agent Security](#) [500+ citations](#) · [300+ GitHub stars](#)
- [2] **Hanrong Zhang**, Shicheng Fan, Henry Peng Zou, Yankai Chen[†], Zhenting Wang, Jiayu Zhou, Chengze Li, Wei-Chieh Huang, Yifei Yao, Kening Zheng, Xue Liu, Xiaoxiao Li, Philip S. Yu, *CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification*, **COLM 2026**. [Code] [RSI](#) [110+ citations](#)
- [3] **Hanrong Zhang**, Zhenting Wang, Tingxu Han, Mingyu Jin, Chenlu Zhan, Mengnan Du, Hongwei Wang[†], Shiqing Ma, *Invisible Backdoor Attack in Self-supervised Learning*, **CVPR 2025**. [Code] [AI Security](#)
- [4] Zhaofen Wu*, **Hanrong Zhang**^{*†}, Fulin Lin, Wujiang Xu, Xinran Xu, Yankai Chen, Henry Peng Zou, Shaowen Chen, Weizhi Zhang, Xue Liu, Philip S. Yu, Hongwei Wang[†], *GAM: Hierarchical Graph-based Agentic Memory for LLM Agents*, **ACL 2026**; ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems [Oral](#). [Agent Memory](#)

- [5] **Hanrong Zhang**, Yankai Chen[†], Shicheng Fan, . . . , Dawn Song, Philip S. Yu, Xue Liu, *Scaling LLM Agent Learning with Data Synthesis: A Comprehensive Survey*, **Preprint**. [Agent Learning & Data Synthesis](#)
- [6] Zhiling Yan, Dingjie Song, **Hanrong Zhang**, Wei Liang, Yuxuan Zhang, Yutong Dai, Lifang He, Philip S. Yu, Ran Xu, Xiang Li, Lichao Sun, *OpenSkill: Open-World Self-Evolution for LLM Agents*, **EMNLP 2026**. [Code] [Agent Self-Evolution](#)
- [7] **Hanrong Zhang**, Yifei Yao, Zixuan Wang, Jiayuan Su, Mengxuan Li, Peng Peng, Hongwei Wang, *Class Incremental Fault Diagnosis under Limited Fault Data via Supervised Contrastive Knowledge Distillation*, **IEEE Transactions on Industrial Informatics**. [Continual Learning](#) [IF 12.3 · SCI Q1 · Top Journal](#)
- [8] Yifei Yao*, **Hanrong Zhang***, Fulin Lin*, Ziyang Jin, Xiaoxiao Li, Hongwei Wang, Ying Chi, *From Uncertainty to Clarity: Uncertainty-Guided Class-Incremental Learning for Limited Biomedical Samples via Semantic Expansion*, **IEEE Journal of Biomedical and Health Informatics**. [Continual Learning](#) [IF 7.7 · SCI Q1](#)
- [9] Xingyue Wang*, **Hanrong Zhang***, Xinlong Qiao, Ke Ma, Shuting Tao, Peng Peng, Hongwei Wang, *Generalized Out-of-distribution Fault Diagnosis (GOOFD) via Internal Contrastive Learning*, **IEEE Transactions on Industrial Informatics**. [OOD Detection](#) [IF 12.3 · SCI Q1 · Top Journal](#)

Other co-authored papers: NeurIPS 2026 ×2, ICML 2026, ACL 2026 ×2, EMNLP 2026 ×3, TMLR, etc.

RESEARCH EXPERIENCE

Agent Security Bench (ASB)

ICLR 2025, First Author

- **Motivation:** LLM agents extend LLM reasoning into real-world action: by calling tools, updating memory, and executing multi-step plans, they directly change external environments, making agent security more critical than standard AIGC safety; yet prior evaluations lacked a unified benchmark spanning prompts, tools, memory, and planning.
- **Contribution:** Introduce the **first fundamental agent security evaluation benchmark**, with 10 scenarios, 10 agents, 400+ tools, 27 attack/defense methods, 7 metrics, and Net Resilient Performance (NRP) for balancing utility and security.
- **Results:** Across 13 LLM backbones and 90,000+ test cases, reveal that current LLM agents remain highly vulnerable, with the highest average attack success rate reaching 84.30%, while existing defenses offer limited protection.

Towards Self-Evolving Skills in LLM Agents

CoEvoSkills (COLM 2026, First Author) & OpenSkill (EMNLP 2026)

- **Motivation:** Agent skills can boost performance on professional tasks, but we identify a **human-machine misalignment**: human-curated skills are costly, and skills humans find helpful are not necessarily useful to agents, sometimes even hurting performance in SkillsBench domains; agents should therefore author their own skills.
- **Contribution:** Propose **CoEvoSkills**, which co-evolves multi-file skill packages with a surrogate verifier in closed-world settings, and **OpenSkill**, which self-evolves skills via open-world knowledge retrieval.
- **Results:** **CoEvoSkills** achieves a 71.1% average pass rate on SkillsBench (+40.5pp over no skills and +17.6pp over human-curated skills); **OpenSkill** reaches 43.6% with Claude Opus 4.6 and 42.1% with GPT 5.2, within 1–3pp of human-curated skills and about 9pp above the best closed-world baseline.

Hierarchical Graph-based Agentic Memory (GAM)

ACL 2026, Co-First & Corresponding Author

- **Motivation:** Long-term agent memory must reconcile fast contextual adaptation with stable knowledge retention: stream-based memories update quickly but lack write isolation, causing memory loss and semantic drift, whereas static graph memories preserve structure but adapt poorly to fluid open-domain dialogue.
- **Contribution:** Propose **GAM**, a hierarchical graph memory framework that decouples memory encoding from consolidation: an Event Progression Graph isolates ongoing dialogue events, and semantic-boundary triggers consolidate completed episodes into a Topic Associative Network for stable long-term knowledge.
- **Results:** **GAM** outperforms six memory baselines on both LoCoMo and LongDialQA.

OPEN-SOURCE PROJECTS

Dr. Claw [Code]

[1.1k+ GitHub stars · 100+ forks](#)

EMNLP 2026 · Second Author · Core Contributor

- Build an **Auto Research** workspace that orchestrates agentic pipelines from research ideation and literature/code survey to experiment execution, analysis, paper writing, and presentation generation.
- Integrate multi-source research news tracking, 100+ research skills, project/session management, and multi-agent backends into a full-stack research IDE.

INVITED TALKS

Towards Self-Evolving AI Agent Skills

Alibaba Group · Jun. 2026

How to Evaluate AI Agent Security?

National University of Singapore · Apr. 2025

ACADEMIC SERVICE

Reviewer ICLR, ICML, NeurIPS, CVPR, ECCV, KDD, AAAI, MICCAI; TPAMI, TNNLS, TMLR, etc.